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INFORMATION: ABILITY TO DISTINGUISH RELIABLY BETWEEN POSSIBLE ALTERNATIVES (PHYSICAL SIGNALS)

- INFORMATION CAN BE TRANSFORMED FROM A PHYSICAL REPRESENTATION TO ANOTHER

CODE: ONE-TO-ONE FUNCTION THAT RELATES THE MESSAGE VALUE TO THE SIGNAL

→ A MESSAGE CAN HAVE MORE THAN ONE REPRESENTATION, ALL ARE EQUIVALENT

FROM A PHYSICAL POINT OF VIEW THEY MIGHT NOT

TO DISTINGUISH YOU NEED AT LEAST TWO POSSIBLE ALTERNATIVES
→ BASIC → A BIT IS THE MINIMUM

BITS: GENERIC TERM FOR A PHYSICAL SYSTEM WITH TWO POSSIBLE DISTINGUISHABLE STATES ⇒ 0/1 ON/OFF TRUE/FALSE

FUNDAMENTAL UNIT OF INFORMATION

1 BYTE = 8 BITS

2 BYTES = 16 BITS

$$2^{16} = 65536$$

$$2^8$$

$$2^{16}$$

ENTROPY: IF M IS THE NUMBER OF POSSIBLE VALUES OF A MESSAGE, WE DEFINE THE ENTROPY OF THE MESSAGE AS

$$H = \log_2(M) \Rightarrow \text{IN THE COURSE } \log_2(x) = \log(x)$$

↳ BECAUSE ARE BITS

- ASSUME THAT ALL THE POSSIBLE VALUES ARE EQUALLY PROBABLE
- THE DEFINITION RESSEMBLES THAT OF THE THERMODYNAMIC ENTROPY $S = K_B \ln(W)$
- THE ENTROPY OF A MESSAGE MEASURES THE INFORMATION NECESSARY TO ENCODE THE MESSAGE
- THE ENTROPY OF A MESSAGE GIVES THE MINIMUM NUMBER OF BITS TO ENCODE A MESSAGE

ES: MESSAGE M CONTAINING AN INTEGER NUMBER FROM 0 TO 9

$$H(M) = \log(10) = 3.32$$

NEED AT LEAST 4 BITS TO ENCODE THIS MESSAGE

IF A AND B ARE TWO INDEPENDENT MESSAGES WITH M_A AND M_B THE NUMBER OF THE POSSIBLE VALUES OF A AND B RESPECTIVELY, THE ENTROPY OF THE JOINT MESSAGE AB IS GIVEN BY

$$H(AB) = H(A) + H(B)$$

THE VALUE THAT MESSAGE A CAN TAKE IS COMPLETELY INDEPENDENT FROM THE VALUE THAT MESSAGE B CAN TAKE

ES:

HOW MANY BITS TO ENCODE AN INTEGER NUMBER FROM 0 TO 999

0 - 999

NUMBERS

$$\log_2(1000) \Rightarrow \frac{\log 1000}{\log 2} = \frac{3}{0.3} = 10$$

NOISE: (AFFECT CAPABILITY TO DISTINGUISH RELIABLE BETWEEN POSSIBLE ALTERNATIVES)

HAVE SOME IMPACT ON THE INFORMATION WE HAVE

ANYTHING THAT PREVENTS A SIGNAL TO BE READ UNAMBIGUOUSLY

→ # OF POSSIBLE MESSAGE VALUE

$$H - H' = \log\left(\frac{M}{M'}\right) \Rightarrow \text{AMOUNT OF INFO. LOST BECAUSE OF THE NOISE}$$

↳ # OF POSSIBLE VALUES THAT CAN BE READ

$$\text{IF } M = M' \Rightarrow \log\left(\frac{M}{M'}\right) = 0 \Rightarrow \text{NO INFO. LOST}$$

Exercise 1.7 A friend is thinking of a number between 1 and 20 (inclusive). She tells you that the number is prime. How much information has she given you?

[1, 20] • PRIME 1, 2, 3, 5, 7, 11, 13, 17, 19 ⇒ 9 PRIMES

$$H = \log(20) \Leftarrow \text{AMOUNT OF INFO. WE NEED}$$

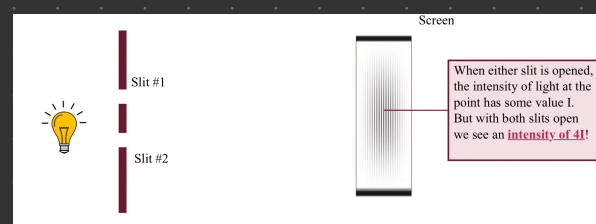
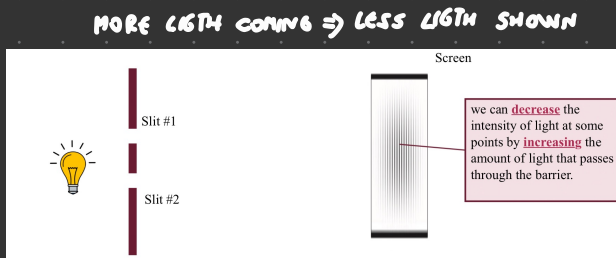
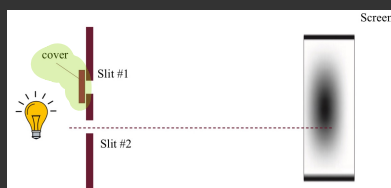
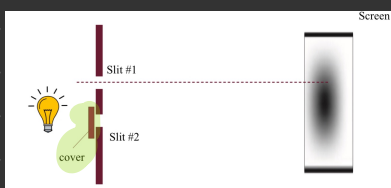
$$H' = \log(9)$$

$$H - H' = \log(20) - \log(9) = \log\left(\frac{20}{9}\right) = 1.27 \quad \leftarrow \text{"A BIT" MORE THAN 1 BIT}$$

WAVES OR PARTICLE? NATURE OF LIGHT

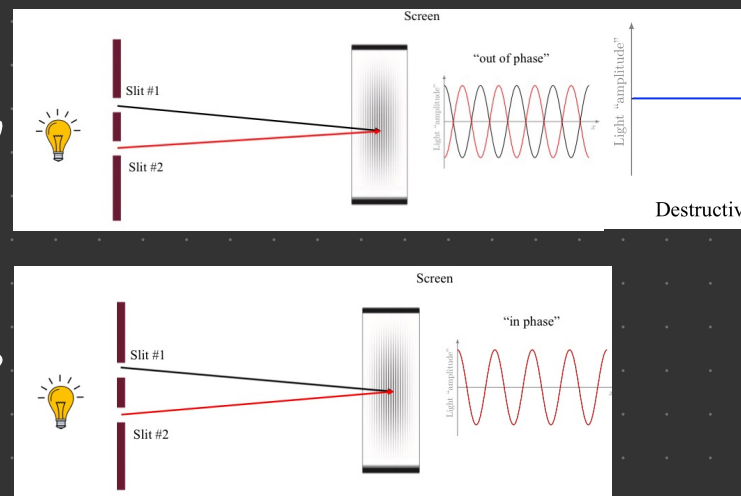
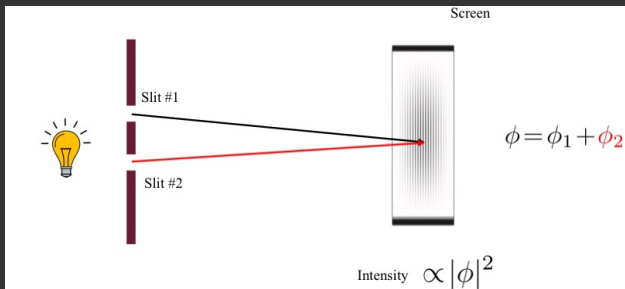
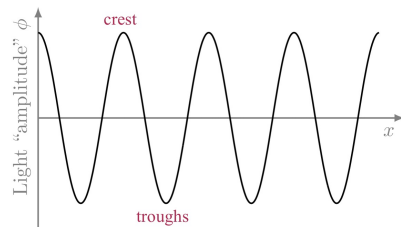
LIGHT FOR HUYGENS: WAVE PHENOMENON

LIGHT FOR NEWTON: PARTICLE-LIKE "CORPUSCLES"



INTERFERENCE:

- The result can be explained if we assume that light behaves as a wave



OUT OF PHASE = RESULT IS 0
IN PHASE = MAXIMUM, SUM TOGETHER THE AMPLITUDE
SQUARE IT $\Rightarrow$ 4 TIMES THE AMPLITUDE

INTENSITY $\propto |\phi|^2$ ^{"PHI"}
 $\downarrow$ PROBABILITY AMPLITUDE
ONLY POSITIVE NUMBER

PLANCK + EINSTEIN $\Rightarrow$ INTERACTION OF LIGHT WITH MATTER

ENERGY OF LIGHT IS CARRIED BY DISCRETE
LIGHT QUANTA LATER CALLED PHOTONS

PHOTONS

ENERGY $E = hf$

MOMENTUM $p = \frac{h}{\lambda}$

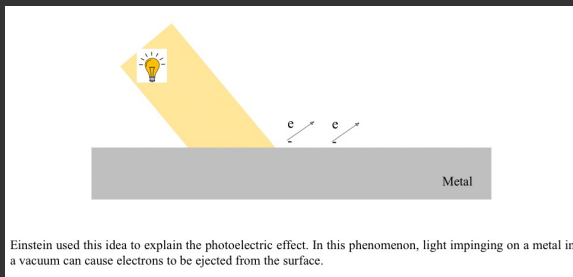
RELATE OF
WAVELENGTH

$$f = \frac{v}{\lambda} \left[\frac{m/s}{m} = \frac{1}{s} = Hz \right]$$

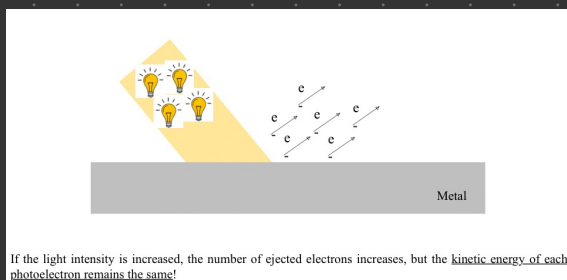
PLANCK CONSTANT $h = 6.626 \times 10^{-34} \text{ Js}$

$$\hbar = \frac{h}{2\pi} = \frac{6.626 \times 10^{-34} \text{ Js}}{6.28} \approx 1.05 \times 10^{-34} \text{ Js}$$

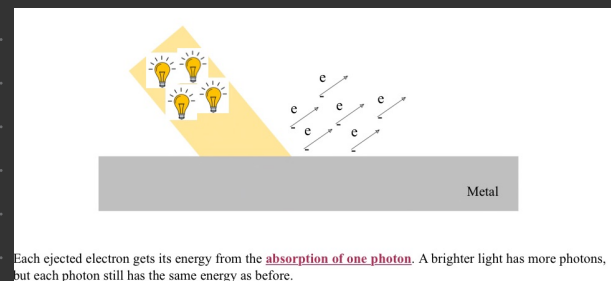
PHOTOELECTRIC EFFECT



Einstein used this idea to explain the photoelectric effect. In this phenomenon, light impinging on a metal in a vacuum can cause electrons to be ejected from the surface.



If the light intensity is increased, the number of ejected electrons increases, but the kinetic energy of each photoelectron remains the same!



Each ejected electron gets its energy from the absorption of one photon. A brighter light has more photons, but each photon still has the same energy as before.

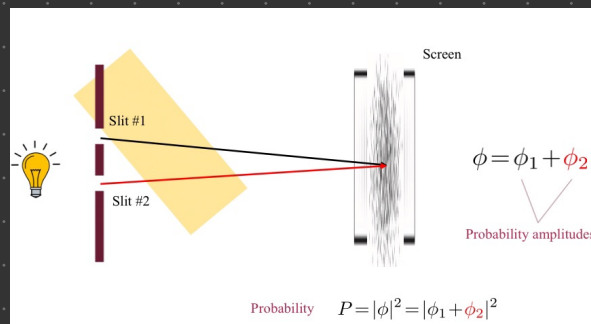
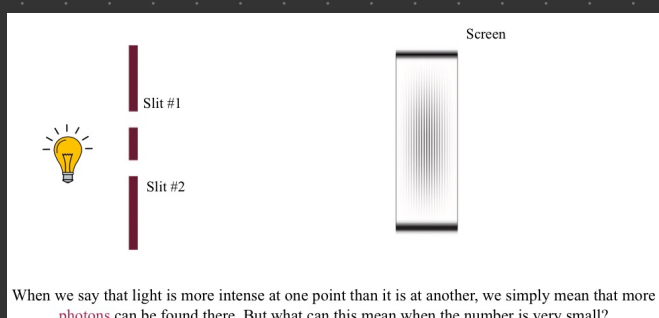
- MORE LIGHT $\Rightarrow$ MORE ELECTRONS
- KINETIC ENERGY OF EACH PHOTOELECTRON REMAINS THE SAME $\Rightarrow$ SAME SPEED

LIGHT IS A WAVE AND A PARTICLE!

- DE BROGLIE $\Rightarrow$ PARTICLES OF MATTER ALSO HAVE WAVE PROPERTIES

AS A GENERAL PRINCIPLE: EVERYTHING IN NATURE CAN BE DESCRIBED IN TERMS OF WAVE AND PARTICLE (WAVE-PARTICLE DUALITY)

SOMETIMES THE WAVE NATURE PREVAILS, SOMETIMES THE PARTICLE NATURE PREVAILS



$$\phi = \phi_1 + \phi_2$$

- PROBABILITY $P = |\phi|^2 = |\phi_1 + \phi_2|^2$
- THE PROBABILITY AMPLITUDE WAVE ϕ OBEYS THE PRINCIPLE OF SUPERPOSITION

QUANTUM MECHANICS PREDICTS ONLY THE PROBABILITY OF AN EVENT, NOT WHETHER OR NOT THAT EVENT WILL DEFINITELY OCCUR

THE TWO PROBABILITY AMPLITUDES INTERFERE, ENHANCING OR SUPPRESSING THE PROBABILITY THAT THE PHOTON LANDS AT A GIVEN POSITION

$$P = |\phi|^2 = |\phi_1 + \phi_2|^2$$



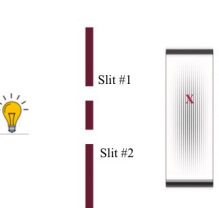
P_1 = probability of having two heads
 P_2 = probability of having two tails

Let's flip a coin twice. We want to calculate
 P = probability of either getting two heads or two tails

Since the two events are mutually exclusive, we now from classical probability theory that

$P = P_1 + P_2$

$$P = \frac{1}{2} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{2}$$



A single photon can either pass through slit #1 or slit #2, the two events are mutually exclusive.

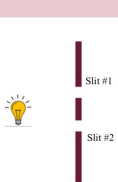
$P_1 = |\phi_1|^2$ = probability that the photon passes through slit #1 and reaches the screen in X
 $P_2 = |\phi_2|^2$ = probability that the photon passes through slit #1 and reaches the screen in X

P = probability that the photon passes either through slit #1 or slit #2 and reaches the screen in X
 $= |\phi_1 + \phi_2|^2 \neq |\phi_1|^2 + |\phi_2|^2$ $P \neq P_1 + P_2$

A strange situation



$P = P_1 + P_2$



$P \neq P_1 + P_2$

How do we know which of these rules will apply in a given situation?